

A. **Diophantine: Solution Existence.** A **Diophantine equation** has form:

$$ax + by = c$$

where a , b , c are integer constants, with a and b not both zero.

The goal is to find integer solutions x and y .

It has an integer pair of solutions if and only if c is:

Therefore, if it has a solution, then:

The Diophantine equation $ax + by = c$ has no solutions if:

Suppose on the other hand that c is a multiple of $\gcd(a, b)$.

$$c =$$

We can reverse the Euclidean algorithm to find integers u and v so:

Thus:

$$c =$$

and thus a solution is given by:

$$x =$$

$$y =$$

The Diophantine equation $ax + by = c$ has a solution if:

Example 1. Find a solution to the Diophantine equation, if possible.

(a) $7544x + 230y = 80$

Solution. Last class we found $\gcd(7544, 230) = 46$.

Since:

this equation has:

(b) $7544x + 230y = 138$

Solution. Once again we have $\gcd(7544, 230) = 46$.

Since:

and thus:

this equation has:

Last class we reversed the Euclidean algorithm to find:

$$46 = -1 \cdot 7544 + 33 \cdot 230$$

$$138 =$$

So a solution to $7544x + 230y = 138$ is given by:

$$x =$$

$$y =$$

Using principles from the next page, the general solution to $7544x + 230y = 138$:

$$x =$$

$$y =$$

where:

B. Diophantine: General Solution. Consider a Diophantine equation.

$$ax + by = c$$

Suppose it has at least one solution: $x = x_0$ and $y = y_0$.

That is to say: $ax_0 + by_0 =$

Let $d = \gcd(a, b)$.

Then the following also solves the Diophantine:

$$x =$$

$$y =$$

as we can see by plugging into equation $ax + by$ and confirming we get c .

In fact, all solutions have this form, as we shall prove.

Let another solution be given: $x = x_1$ and $y = y_1$.

That is to say: $ax_1 + by_1 =$

Subtract $ax_0 + by_0 = c$ from this to obtain:

Divide both sides by the gcd d and rearrange to obtain:

It follows that b/d divides:

But since $d = \gcd(a, b)$ we have:

So, by Euclid's lemma, in fact:

Thus: $x_1 - x_0 =$

Substitute this into the equation $a(x_1 - x_0) + b(y_1 - y_0) = 0$ to obtain:

We summarize below.

Consider a Diophantine $ax + by = c$ having a solution $x = x_0$ and $y = y_0$.

If $d = \gcd(a, b)$ then the general solution is:

$$x =$$

$$y =$$

where t is an arbitrary integer.

Example 2. A customer bought a dozen pieces total of apples and bananas, at least one of each, for a total of \$1.32. If an apple costs 3 cents more than a banana, how many pieces of each kind were bought, and for what prices?

Solution. Let:

$$m =$$

$$n =$$

$$p =$$

$$q =$$

These are all positive integers and:

$$132 =$$

$$12 =$$

$$p =$$

Substitute the second and third equations into the first.

This is the Diophantine equation $ax + by = c$ we will solve.

Let $d = \gcd(a, b) =$

Write $\gcd(a, b)$ as an int. combo of a and b — either Euclidean algorithm or guess.

$$32 =$$

So we have found a solution to $32 = 4q - 1n$:

$$q =$$

$$n =$$

The general solution determined the rule $x = x_0 + (b/d)t$ and $y = y_0 - (a/d)t$ is:

$$q =$$

$$n =$$

$$m =$$

$$p =$$

The choices of t that yield positive integers are: $t =$

The corresponding solutions are:

$$q = \quad n = \quad m = \quad p =$$

$$q = \quad n = \quad m = \quad p =$$